GY462—Real Estate Finance

# Lecture 4

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## Leverage and Yield

- **WACC formula can be applied to any additive component of total return** $r$
- $r = y + g$ ; $r = r_f + RP$
- **Components**
  - **income yield** $y$
  - **growth** $g$ *(property value appreciation)*
  - **risk premium** $RP$

⇨ **…**



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## Example: Japanese Property

- **Buy £100 million of Japanese RE at cap rate of 7%**
- **Assume falling net rents ~ -2% p.a. (in 2006) ⇨ Justification?**
  - **No noteworthy (nominal) economic growth over last 13 yrs**
  - **Real estate supply growth ~+2% p.a.**

| Year | 2006 | 2007 | 2008 | 2009 |
|---|---|---|---|---|
| NOI | 7 | 6.86 | 6.72 | 6.59 |



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## Example: Japanese Property — *Continued*

- **Further assumptions**
  - **LTV = 85%**
  - **Interest rate = 4% (!)**
  - **Non amortising loan ⇨ interest payments p.a. = 3.4**

⇨ **<u>Cash-on-cash yield</u>?**



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## Japanese Property
## *Cash-on-Cash Yield*

- **Cash-on-cash (current) yield for 2006:**

$$y_E = \frac{(7\% - 0.85 \times 4\%)}{(0.15)} = 24\%$$

- **Cash-on-cash yield for 2009: 21.3%**
  - **Lower NOI from leasing property**

⇨ **Problem?**





## Japanese Property
## *– Property Value*

- **Declining property value (as NOI falls)**
- **Assume 7% going out cap rate**

$$V_{2009} = \frac{(6.59 \times (1 - 0.02)}{0.07} = 92.26$$

- **Equity investment left in 2009: 92.26 – 85 = 7.26 (of original 15)**





## Japanese Property
## *– Equity Return from Appreciation*

- **Equity return from capital appreciation:**

$$g_E = \frac{(g_P - LTV \times g_D)}{(1 - LTV)} = \frac{(-7.74\% - 0.85 \times 0)}{(0.15)} = -51.6\%$$

*or alternative calculation:*

$$g_E = \frac{7.26 - 15}{15} = \frac{-7.74}{15} = -51.6\%$$

- **Annual expected equity capital gain: -16.6%**





## Example: Japanese Property
## *—Conclusions*

- **Example demonstrates dilemma of Japanese market for property investors**
  - **Attractive cash-on-cash yields from positive leverage**
  - **BUT… declining values!**
- **If economy stabilises or rebounds ⇨ 'Homerun'**
- **If market becomes unexpectedly worse or cap rates rise slightly ⇨ 'Disaster'**

⇨ **<u>Leverage and risk</u> …**





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## Leverage and Risk — *Example*

- **Consider RE project**
  - **Total cost (in t=0): 1**
  - **Holding period = 5 years (t=1,…,5)**
  - **Discount rate = 10%**
  - **3 possible scenarios**
    - **Pessimistic:** $NOI_1 = 0.1$ ; $g = -2\%$ ; $V_S=0.9$ ; $p=0.2$
    - **Most likely:** $NOI_1 = 0.11$ ; $g = 0\%$ ; $V_S=1.1$; $p=0.5$
    - **Optimistic:** $NOI_1 = 0.12$ ; $g = +5\%$ ; $V_S=1.3$ ; $p=0.3$



## Risk Level for Unlevered Investment

| UNLEVERED | (1) Estimated $BTIRR_P$ | (2) Expected $BTIRR_P$* | (3) Deviation | (4) Squared Deviation | (5) Probability | (6) Product (4) * (5) |
|---|---|---|---|---|---|---|
| Pessimistic | 7.93 | 13.06 | -5.13 | 26.31 | 0.2 | 5.26 |
| Likely | 12.56 | 13.06 | -0.50 | 0.25 | 0.5 | 0.12 |
| Optimistic | 17.31 | 13.06 | +4.25 | 18.07 | 0.3 | 5.42 |

*Expected $BTIRR_P$ = (0.2)*7.93 + (0.5)*12.56 + (0.3)*17.31 = 13.06

$\Rightarrow$ Variance $= 5.26 + 0.12 + 5.42 = 10.81$

$\Rightarrow$ Standard Deviation $= \sqrt{10.81} = 3.29$



## Risk Level for Levered Investment (Using Fully Amortizing CPM)

| LEVERED *(Loan = 0.9, interest rate = 10%, 15-year term, annual compounding)* | (1) Estimated $BTIRR_E$ | (2) Expected $BTIRR_E$* | (3) Deviation | (4) Squared Deviation | (5) Probability | (6) Product (4) * (5) |
|---|---|---|---|---|---|---|
| Pessimistic | -6.03 | 25.46 | -31.50 | 991.95 | 0.2 | 198.39 |
| Likely | 24.96 | 25.46 | -0.51 | 0.26 | 0.5 | 0.13 |
| Optimistic | 47.30 | 25.46 | +21.84 | 476.98 | 0.3 | 143.09 |

*Expected $BTIRR_E$ = (0.2)*-6.03 + (0.5)*24.96 + (0.3)*47.30 = 25.46

$\Rightarrow$ Variance $= 198.39 + 0.13 + 143.09 = 341.61$

$\Rightarrow$ Standard Deviation $= \sqrt{341.61} = 18.48$



## Formalizing the Relationship between Leverage and Risk

- **Recall** $r = r_f + RP$
- **So applying WACC formula**

  $RP_P = (LTV)\ RP_D + (1 - LTV)\ RP_E$
- **Reformulate to …**

  $RP_E = RP_D + LR\ (RP_P - RP_D)$
- **If debt is riskless then**

  $RP_E = LR \times RP_P$

  *($RP_E$ is **directly** proportional to LR!)*



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## Conclusions

- ⇨ ***Higher risk*** is implicit cost associated with financial leverage!
  - No free lunch in the use of leverage!
- ⇨ Always ask…
  - Is higher expected return commensurate with higher risk?
  - Ways to realize higher return with less risk?



## This Lecture

- Project development
  - Acquire land and develop real estate asset
  - Key insights and simple metrics
- Development and project financing
  - Different loan structures
  - Calculation of project returns



## Project development — *What do we mean?*

- So far, we have focussed on property acquisition and management
  - Buy, carry out 'light touch' redevelopment
  - Let out to tenants and sell
- But a lot of real estate investment is about development
  - Buy a plot of land and build an income generating property
- How do we go about thinking and modelling such an investment?



## Important considerations

- Project development = process of financing acquisition of land with the intent of…
  - Building
  - Leasing
  - Managing
  - Selling
- Very complex endeavour requiring wide ranging skills and contingencies



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## Project development in steps
## — *Lecture key elements*

- Planning and permitting process
- Risks and feasibility considerations
- Development finance
  - Construction lending
  - Permanent loan
- A simplified investment case – 'short-cut'
- A fully-fledged project development case



Step 1

## Planning and permitting in project development



## What to consider?

- Project development usually begins with
  - An application that identify a site where to build
  - A preliminary project that is reviewed by public officials
- The set of specific information required varies depending on institutional context
  - But planning regulations are a key element of project development
  - Need to understand the institutional background



## Key planning concepts
## – *Part I*

- Setbacks and right-of-way line
- Footprint, façade, bulk, building materials
- Floor-to-Area (FAR) ratios, height restrictions, views preservation
- Inclusionary-zoning, minimum-lot size, mixed-use development



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## Key planning concepts – *Part II*

- Planning permits and restrictions will vary significantly
  - Across countries and jurisdictions
  - Even within countries, as public officials might have a reputation for being 'strict' or 'permissive'
  - This is very much the case in the UK system where Local Authorities have margins of discretion
- Taxonomy of planning concepts likely to vary…
- But always refers to elements of building itself and in relation to the neighbourhood in which it sits



Step 2

## Project development – risk and feasibility



## The development timeline – *how long?*

- Essentially a developer buys, develops, lets out, manages and sells
- This takes several years, and exact timeline depends on the business strategies
  - Long-term approach: own and manage many years after construction
  - Short/Mid-term: sell project to institutional investor during lease-up stage / as soon as leased
  - Build as part of bigger master-planned development – in business and industrial parks



## Different timelines, different risks

- Different parts of the timeline entail different levels of risk
  - This peaks during construction and completion / occupancy phase
  - Different developer can choose to specialise
    - Cover the whole project → have the ability to deal with all the risks
    - Short/mid-term strategy → outsource phases to external contractors who can help manage risk
- Risk / viability depends on far future demand for space – must 'know' market very well



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## Important risks and cost factors to bear in mind

- Construction timeline very long: CBD office / high-rise resi.– 2-4 years; shopping centre–3-6 years
- Hard-costs – like an S-shape (low, peak, flat)
- Soft-cost?
  - Loan interests: how can you pay when you have no income, only costs? Negative amortization
  - Other costs: project management, back office, etc.
- Forecasting is key – including how to manage contractors
  - If business is 'good', will they take too much on board, cause delay and make business 'bad'?



## Negative and positive cashflow business

- Development is all about offsetting negative initial cashflows with positive future cashflows
  - But initial negative outlays are clearly less uncertain than future sources of revenues
- Stated differently, initial part of the cash flow should attract a lower discount factor
  - Another way to look at it is to think: "How much money do I need to set aside to be 100% sure I can cover development costs?"
  - Liquid investment rate should be used
- How to offset risks? Pre-letting strategy – comes at a cost: what is the market is better than expected?



Step 3

## Key elements of development finance



## Different funding vehicles

- Land is normally acquired with equity or mostly equity
- A construction or interim loan is then used to cover soft and hard cost of development
- A permanent loan which is used after the development phase is completed to continue manage the property
- Construction and permanent loan can come from two different sources
  - Short/mid-term developer might need only development loan if they plan to sell at/after lease-up phase



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## Construction loan features

- Usually extended by a specialist – know the market, understand the process, can monitor progress
- Disbursement of the funds to develop happens progressively
  - The principle is: never extend funds exceeding the value of the property developed at that stage
  - I.e., the 'collateral' against which the loan is secured
- Most common method of draw-down: monthly
  - Each month draws down on the basis of the value of the work completed the previous month



## Permanent loan features

- Submission for requesting a permanent loan is a complex endeavour
  - Because the permanent lender 'agrees' to pay off developer loan at the end of construction phase
  - And to continue financing the project
- Information submission includes:
  - Project and market information
  - Government and regulatory information
  - Architectural and building costs breakdown
  - Details about the interim lending loan structures



## Permanent loan – *Commitment and contingencies*

- When all documents provided to satisfaction, the permanent lender issues an intent to fund
  - More detailed planning (also required by the interim lender) then trigger a commitment to fund
- The permanent lender specifies some contingencies
  - If these are met, the permanent loan is binding
  - At end of construction, permanent lender provides fund to repay construction loan and carry on with plans
  - If contingencies are not met, no legal obligation for permanent lender
    - Developed either finds another permanent lender or could face foreclosure at the end of the interim loan



## Permanent loan contingencies

- Common take-out contingencies for permanent loan
  - Maximum period of time allowed to secure interim loan
  - Completion date for construction phase
  - Minimum rent-up / leasing requirements
  - Provisions for gaps if rent-up not as planned
  - Approval of design / material changes
- Permanent lenders normally insurance companies, pension funds, etc.
  - Contingencies to guarantee that developers' incentives align with the lender objectives and need for security



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Step 4

## 'Short-cut' valuation of a development project

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## 'Development can pay off handsomely' – *Linneman*

- Development is risk but can be lucrative – how do you know?
- The 'build to a 10, sell to an 8' rule – what does it mean?
  - Note: this means build at X, sell at Y; with X>Y
- Recall cap rates and stabilised properties
  - Stabilised = good occupancy given market and the characteristics of the project



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## Build at X, sell at Y – *In practice*

- Build at 10 means projected stabilised NOI is 10% of projected development cost
  - Essentially a going in cap rate
  - E.g., NOI = £1mio; all in costs = £10mio
- Sell at 8 means projected stablised NOI is 8% of the sale value
  - A going out cap rate
  - E.g., NOI = £1mio; sale price = £12.5mio – with £2.5mio expected value creation
  - Or 25% of the projected costs – handsome return…



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## A simple example (Linneman)

- A developer wants to build a 10 and sell an 8-8.5 – feeling like it needs a 15-20% return

- The Muk Office plaza data are summarised in the table
- The relevant question to hit the 10/8 rule are…

1. What NOI to expect?
2. What costs?
3. And what sale price?

| Muk Office Plaza | |
|---|---|
| Land | £30 per gross foot |
| Hard costs | £90 per gross foot |
| Soft costs | £30 per gross foot |
| *Total costs* | £150 per gross foot |
| Rent | £30 per leasable foot |
| Operating costs | £10 per gross foot |
| Occupancy | 95% of leasable foot |
| Loss factor | 30% of gross footage |



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## The 10/8 rule – *calculations*

- **What stabilised NOI? £30 per foot – but with 95% occupancy = £28.5**
  - Assumes £0 on vacant space – but could put to productive use
  - But must consider some £10 of operating costs to be paid if space is vacant (utilities, insurance, taxes…)
- **This is per gross foot – but must consider leasable space: 1-loss factor at 20%**
  - £28.5 * 70% - £10 = £9.95; considering £150 costs = 6.6%
  - If your rule is '10', 6.6% is too low to compensate for the risk implied by taking on development
- **How high should the rent be to justify development?**



## The 10/8 rule – *calculating the replacement rent*

- **To go in at 10%, you need an NOI £15 given total gross costs of £150**
  - That implies a market rent of £25 – how to get that?
  - 10% of £150 + the £10 of operating costs
- **What is that in terms of rent per leasable foot? Must take into account vacancy and loss factor**
  - £25 * (1/(1-Loss Factor)) * (1/(1-Vacancy))
  - £25 * (1/0.7) * (1/0.95) = £37.59
  - That is 25% higher than current estimates – or 3% per year over 7.6 years. Can that be achieved? Maybe…
  - Calculations using '10/8' rule gives you a quick way to determine feasibility
  - PS – what about the '8'? *Discuss*



Step 5

## A fully fledged development case



## A fully-fledged development example (B&F, Ch. 16)

- **Developing Rolling Meadows Centre**
- **Plans implemented by Southfork Development – expects to own and operate for a long time**
- **Expects to use interim and permanent finance**
- **Citadel Life Insurance has been approached to provide permanent funding**
- **What is the projected return on this development? A long way of intermediate steps to get there…**



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## Project description (Exhibit 16-3)

| A. Site + improvements | | D. Loan Information | |
|---|---|---|---|
| Site area (acres) | 9.5 | ***Construction loan*** | |
| Gross Buildable area | 120,000 sq.ft. | Term | 12 months |
| Gross leasable area | 110,000 sq.ft. | % drawn first 4 months | 75% |
| % leasable | 91.67% | % drawn last months | 25% |
| FAR (site area) | 3.45 | Interest | 12% |
| Parking spaces | 550 | Loan Fee | 2% |
| **B. Development period** | **12 months** | ***Permanent loan*** | |
| **C. Site Plan** | | Amortization period | 25 years |
| Building coverage | 29% | Term of loan | 10 years |
| Street parking | 45% | Interest | 12% |
| Open Space | 26% | Loan fee | 3% |
| | | **E. Anticipated hold after construction** | **5 years** |

- **Notice permanent loan has amortizing features!**



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## Cost Information (Exhibit 16-4) – *Part I*

| A. Land/ site improvement | | Project management fee | £300,250 |
|---|---|---|---|
| ***Site acquisition*** | £2,500,000 | Finish out | £1,400,600 |
| ***Improvements*** | | Graphics | £66,570 |
| Off site | £250,000 | ***Total hard costs*** | ***£7,055,500*** |
| Excavation | £50,000 | ***Soft costs*** | |
| Sewer/water | £150,000 | Architect ** | £147,000 |
| Paving | £200,000 | Fees and permits ** | £24,300 |
| Sidewalks | £100,000 | Legal fees ** | £26,900 |
| Landscaping | £100,000 | Construction interests | £692,416 |
| ***Total improvements*** | ***£850,000*** | Construction loan fees ^^ | £180,028 |
| **B. Construction costs** | | Permanent loan fees ^^ | £270,042 |
| ***Hard costs*** | | Leasing commissions ** | £45,300 |
| Shell structure | £3,925,000 | Direct overheads ** | £160,000 |
| HVC | £528,500 | Indirect overheads ^^ | £30,800 |
| Electrical | £613,000 | ***Total soft costs*** | ***£1,576,787*** |
| Plumbing | £221,580 | ***Total project costs*** | ***£11,982,287*** |



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## Cost Information (Exhibit 16-4) – *Part II*

| Construction loan request | |
|---|---|
| Total improvements | £850,000 |
| Total hard costs | £7,055,500 |
| Soft costs ** | £403,500 |
| Total cost to be financed | £8,309,000 |
| Estimated interest carry | £692,416 |
| ***Total loan amount*** | ***£9,001,416*** |
| ***Equity requirements*** | ***£2,980,871*** |
| **Total project costs** | **£11,982,287** |

- **Land purchase normally funded with equity**
- **Not all soft costs are covered by the loan**
- **Some fees considered 'intangible' that cannot be recovered in case of default**
  - **In our example ** are included but ^^ are not included**
- **An estimate of the construction interest is included in the loan request**

- **Note that lender might expect a loan-to-appraised value of at most (say) 80%. That means equity contribution (land and fees) should be at least 20%**
- **If the value of the improvements is above that 80% (say 90%), more equity will have to be injected**



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## Project loan schedule (Exhibit 16-5)

| | Loan Draws | | | | Payments | | |
|---|---|---|---|---|---|---|---|
| Month | Costs | Interest | Balance | | Interest | Amort. | End. balance |
| 1 | 1,557,938 | 0 | 1,557,938 | | | | 1,557,938 |
| 2 | 1,557,938 | 15,579 | 3,131,454 | | 15,579 | 0 | 3,131,454 |
| 3 | 1,557,938 | 31,315 | 4,720,706 | | 31,315 | 0 | 4,720,706 |
| 4 | 259,656 | 47,207 | ,8516,325 | | 47,207 | 0 | ,8516,325 |
| 5 | 259,656 | 63,259 | 6,648,766 | | 63,259 | 0 | 6,648,766 |
| 6 | 259,656 | 66,488 | 6,974,910 | | 66,488 | 0 | 6,974,910 |
| 7 | 259,656 | 69,749 | 7,304,315 | | 69,749 | 0 | 7,304,315 |
| 8 | 259,656 | 73,043 | 7,637,014 | | 73,043 | 0 | 7,637,014 |
| 9 | 259,656 | 76,370 | 7,973,041 | | 76,370 | 0 | 7,973,041 |
| 10 | 259,656 | 79,730 | 8,312,427 | | 79,730 | 0 | 8,312,427 |
| 11 | 259,656 | 83,124 | 8,655,208 | | 83,124 | 0 | 8,655,208 |
| 12 | 259,656 | 86,552 | 9,001,416 | | 86,552 | 9,001,416 | 0 |
| **Total** | **8,309,000** | **692,416** | | | | | |

- **First 4months: 75% of costs; remaining 8months: 25% of costs**



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## Some considerations and permanent loan (Exhibit 16-6)

| Permanent loan | |
|---|---|
| Total loan | £9,001,416 |
| Debt amortization | 25 |
| Term of loan | 10 |
| Interest rate | 12% |
| Debt service/year | 1,137,661 |
| 3% loan fee | £270,042 |
| *Yield to lender* | **12.55%** |

- **The construction loan is priced at 12%/12 as has monthly draws**
- **The interest is borrowed – not paid; it accrues and is part of the principal**
- **Meaning: no payment to lender till end of the loan**
- **Loan based on estimated costs. If actual costs above, lender *not* obliged to meet those**

- **Permanent lender also does not need to commit to more than initial projection – developer needs to plan carefully to avoid foreclosure**
- **Permanent loan is amortizing with monthly compounding – pre-paid at the end of holding period. With origination fee → yield to lender 12.55% (*clarify*)**



## What next?

- **So far focussed on costs and lending side**
- **But in order to close the case, we must formalise the revenues too…**
- **First thing would be a detailed stacking plan / market survey data**
  - **Spells out which tenants (stores in this case) you are expecting to have**
  - **How much space they occupy and how that compares to other 'representative' (relevant markets) – *see book***
- **From this, you can start putting together your pro-forma for holding period**



## Cash Flows Holding Period (Exhibit 16-5) – *Part I*

| Income | Y2 | Y3 | Y4 | Y5 | Y6 |
|---|---|---|---|---|---|
| Minimum rent | 1,650,000 | 1,749,000 | 1,853,940 | 1,965,176 | 2,083,07 |
| Overage (5% sales) | 30,000 | 124,800 | 129,792 | 134,984 | 140,383 |
| *Tenant reimbursements* | | | | | |
| Real estate taxes | 137,000 | 143,000 | 148,720 | 154,669 | 160,856 |
| Common area maint. | 385,000 | 400,400 | 416,416 | 433,073 | 450,396 |
| Utilities | 367,500 | 382,200 | 397,488 | 413,388 | 429,923 |
| Insurance | 33,000 | 34320 | 35,693 | 37,121 | 38,605 |
| Gross potential income | 2,063,000 | 2,833,720 | 2,982,720 | 3,138,410 | 3,303,249 |
| Vacancy Allowance | 789,900 (30%) | 141,686 (5%) | 149,102 (5%) | 156,920 (5%) | 165,162 (5%) |
| **Expected gross income** | **1,822,100** | **2,692,034** | **2,832,947** | **2,981,490** | **3,138,087** |

- **Occupancy: 70% going up to 95%;**
- **Base rent: £15 psf, going up by 6% per year + overage provision (*remember?*);**
- **Tenant reimbursements are negotiable**



## Cash Flows Holding Period (Exhibit 16-5) – *Part II*

| Expenses | Y2 | Y3 | Y4 | Y5 | Y6 |
|---|---|---|---|---|---|
| Manag. and leasing fees | 104,500 | 93,690 | 99,187 | 105,008 | 111,174 |
| General/admin costs | 77,000 | 80,080 | 83,283 | 86,615 | 90,079 |
| Real estate taxes | 137,000 | 143,000 | 148,720 | 154,669 | 160,856 |
| Common area maintenance | 385,000 | 400,400 | 416,416 | 433,073 | 450,396 |
| Utilities | 367,500 | 382,200 | 397,488 | 413,388 | 429,923 |
| Insurance | 33,000 | 34320 | 35,693 | 37,121 | 38,605 |
| Other | 27,500 | 28,600 | 29,744 | 30,934 | 32,171 |
| *Total expenses* | *1,064,800* | *1,092,402* | *1,137,847* | *1,185,215* | *1,234,589* |
| **NOI** | **757,300** | **1,599,632** | **1,695,099** | **1,796,275** | **1,903,498** |
| Less debt (perm. Loan) | 1,137,661 | 1,137,661 | 1,137,661 | 1,137,661 | 1,137,661 |
| **Cash Flow** | **--380,361** | **461,971** | **557,438** | **658,614** | **765,837** |

- **Leases are *net to tenants* – pass through of insurance, property taxes, etc.**
- **Tenants also pay for expenditures on common areas and utilities**
- **The developer then uses these to offset costs**



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## Some simple project metrics

- Calculated from Year 3 – after lease up period when the building is 'normal' (stabilised)
- Total annual expenses / EGI (effective gross rent): approx. 40% - share of income going to expenses
- Debt-coverage-ratio (NOI / debt service) : approx. 1.4
- Free and clear return (NOI / project cost): 13-16%
- RoE (cash flow / equity investment): 15-26% (!!!)



## Profitability and IRR (Exhibit 16-10)

| | Y0 | Y1 | Y2 | Y3 | Y4 | Y5 | Y6 |
|---|---|---|---|---|---|---|---|
| Equity | -2,950,071 | -30,800 | | | | | |
| CF operations | | | -380,361 | 461,971 | 557,438 | 658,614 | 765,837 |
| CF sale | | | | | | | 7,104,160 |
| Total CF | -2,950,071 | -30,800 | -380,361 | 461,971 | 557,438 | 658,614 | 7,869,997 |
| | | | | | | | |
| ***IRR*** | ***21.33%*** | | | | | | |
| ***NPV @ 21%*** | ***38,884*** | | | | | | |

- Year 0: equity to buy land; Year 1: development phase – but indirect overheads (soft costs) not covered by interim loan
- Sale price? 16,035,003 – based on total project cost of 11,982,287 compounded at an appreciation of 6% per year for five year (same as rents)
- Implied cap rate? Roughly 12% - *Discuss*
- CF sale: sale price – mortgage balance (8,610,143; amortizing loan) – selling expenses (320,700)
- NPV at 21% - why? 9p.p. over cost of loan (12%) to compensate for risk



## Concluding remarks

- Project development complex and risky – but appealing remuneration
- Progressively more important for RE
- Next week:
  - Private Equity in RE (pre-recorded)
  - Funds external lecture



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